Appendix B: Extended Methods

B.1 Cox Proportional Hazards Model

Model Specification. The primary model is the Cox proportional hazards regression:

$$h(t|\mathbf{X}) = h_0(t)\exp(\mathbf{X}\beta)$$

where:

- $h(t|\mathbf{X})$ is the hazard of completion at time t given covariates $\mathbf{X}$
- $h_0(t)$ is the unspecified baseline hazard function
- β is the vector of regression coefficients

The semi-parametric nature of the Cox model allows estimation without specifying the baseline hazard, reducing model misspecification risk.

Proportional Hazards Assumption. The Cox model assumes that hazard ratios are constant over time:

$$\frac{h(t|X_1)}{h(t|X_2)} = \exp(\beta(X_1 - X_2)) \quad \forall t$$

We test this assumption using Schoenfeld residuals. For each covariate, we regress scaled Schoenfeld residuals on time functions and test for non-zero slopes. Non-significant tests ($p > 0.05$) support the proportional hazards assumption.

Partial Likelihood Estimation. Parameters are estimated by maximizing the partial likelihood:

$$L(\beta) = \prod_{i:\delta_i=1} \frac{\exp(\mathbf{X}_i\beta)}{\sum_{j\in R(t_i)} \exp(\mathbf{X}_j\beta)}$$

where:

- $\delta_i = 1$ indicates an observed event (completion)
- $R(t_i)$ is the risk set at time t_i (all subjects still at risk)

Standard errors are computed from the observed information matrix.

Hazard Ratio Interpretation. Hazard ratios (HR) quantify the multiplicative change in completion hazard per unit change in the covariate:

- HR > 1: Higher velocity associated with faster completion
- HR < 1: Higher velocity associated with slower completion
- HR = 1: No association

For velocity measured in percentage points per quarter, HR = 4.60 means each additional percentage point of quarterly velocity is associated with a 4.60-fold increase in the instantaneous completion hazard.

B.2 Stratified Analysis

Approach. Context-specific effects are estimated by fitting separate Cox models within subgroups:

$$h_k(t|\mathbf{X}) = h_{0k}(t)\exp(\mathbf{X}\beta_k) \quad \text{for } k = 1, \dots, K$$

where k indexes strata (disaster types, program types, experience levels).

Stratification Variables.

Variable	Categories
Disaster Type	Hurricane, Wildfire, Flood, Other
Program Type	Housing, Infrastructure, Administration
Experience	Novice, Experienced
Program Phase	Early, Middle, Late

Small Sample Considerations. Stratified analyses reduce sample sizes, potentially compromising statistical power. We report confidence intervals alongside p-values to convey uncertainty. Wide confidence intervals (e.g., HR=51.09, 95% CI: 4.01-650.90 for wildfire) indicate effect magnitude is estimated imprecisely even when the direction is statistically significant.

B.3 Interaction Models

Specification. Formal tests of heterogeneity use interaction terms:

$$\log h(t) = \log h_0(t) + \beta_1 \text{Velocity} + \beta_2 \text{Context} + \beta_3 (\text{Velocity} \times \text{Context})$$

The interaction coefficient β_3 tests whether velocity effects differ significantly across contexts. Wald tests assess $H_0 : \beta_3 = 0$.

Power Considerations. Interaction tests require larger samples than main effect tests. With N=152 and substantial censoring, power to detect interaction effects is limited. Non-significant interactions should be interpreted cautiously; they may reflect insufficient power rather than true homogeneity.

B.4 Trajectory Clustering

K-Means Algorithm. Velocity trajectories are clustered using K-means on standardized quarterly velocity profiles:

1. **Standardization**: Each grantee's velocity time series is z-scored to remove scale differences
2. **Alignment**: Series are interpolated to equal lengths (20 time points)
3. **Clustering**: K-means minimizes within-cluster sum of squared distances:

$$\min_{\mathbf{C}} \sum_{k=1}^{K} \sum_{i \in C_k} \|\mathbf{v}_i - \mu_k\|^2$$

where $\mathbf{v}_i$ is grantee i's velocity trajectory and μ_k is cluster k's centroid.

Cluster Selection. The number of clusters (K=3) is selected using:

1. **Elbow method**: Plotting within-cluster sum of squares against K

2. **Silhouette scores**: Measuring cluster separation

3. **Interpretability**: Ensuring clusters have substantive meaning

Cluster Characterization. Clusters are characterized by:

- Mean velocity level (fast, moderate, slow)
- Velocity variance (consistent vs. variable)
- Trajectory shape (accelerating, decelerating, flat)
- Completion outcomes (median completion time, completion rate)

B.5 Phase-Specific Velocity Models

Phase Definition. Programs are divided into terciles based on observed duration:

$$\text{Phase}_t = \begin{cases} \text{Early} & \text{if } t \leq T/3 \\ \text{Middle} & \text{if } T/3 < t \leq 2T/3 \\ \text{Late} & \text{if } t > 2T/3 \end{cases}$$

Multivariate Model. Phase-specific effects are estimated by including all three phase velocities simultaneously:

$$\log h(t) = \log h_0(t) + \beta_E \text{Velocity}_{\text{Early}} + \beta_M \text{Velocity}_{\text{Mid}} + \beta_L \text{Velocity}_{\text{Late}}$$

This specification identifies the independent contribution of each phase, controlling for the others.

Sensitivity Analysis. Alternative phase definitions are tested:

- **Quartiles**: Four equal phases
- **Fixed intervals**: Years 1-3, 4-7, 8+
- **Milestone-based**: Pre-50%, 50-80%, 80-95%

Results are qualitatively similar across specifications.

B.6 Meta-Analysis of Velocity Estimates

Estimate Extraction. From each stratified model, we extract:

- Point estimate: $\hat{\beta}$ (log hazard ratio)
- Standard error: $\mathrm{SE}(\hat{\beta})$
- Sample size: n_k
- Number of events: d_k

Summary Statistics. Rather than pooled estimation (which would require independence assumptions), we compute:

- **Median HR**: Central tendency of effect estimates
- **Range**: Minimum to maximum HR
- **Interquartile range**: Middle 50% of estimates
- **Percent significant**: Proportion with $p < 0.05$
- **Percent accelerating**: Proportion with HR > 1

Forest Plot. Visual display shows each estimate with 95% confidence intervals on a log scale, organized by analysis type. Reference line at HR=1 indicates null effect.

B.7 Sensitivity Analyses

Threshold Sensitivity. Completion thresholds other than 95% are tested:

Threshold	N Events	Median HR	p-value Range
90%	58	4.21	0.008-0.156
95%	40	4.60	0.002-0.891
99%	25	5.12	0.001-0.743

Results are robust to threshold choice.

QA Flag Exclusion. Sensitivity analyses excluding observations flagged for data quality issues:

Exclusion Criterion	N Excluded	HR Change
Extreme velocity	3	+4.2%
Large obligation adjustment	12	-6.1%
Negative expenditure	8	+2.8%
Any flag	18	-3.4%

Results are substantively unchanged by exclusions.

Winsorization Thresholds. Alternative winsorization cutoffs:

Cutoff	Median HR	95% CI
1%/99% (baseline)	4.60	1.42-14.89

Cutoff	Median HR	95% CI
2.5%/97.5%	4.43	1.38-14.21
5%/95%	4.28	1.31-13.98

Effects are modestly attenuated with stricter winsorization but remain statistically significant.

B.8 Software and Reproducibility

Computational Environment.

- **Python 3.11** with lifelines package for survival analysis
- **scikit-learn** for K-means clustering
- **pandas/numpy** for data manipulation
- **matplotlib/seaborn** for visualization

Code Availability. Analysis scripts are available in the project repository:

- `run_phase_specific_analysis.py`: Phase velocity models
- `run_trajectory_clustering.py`: K-means clustering
- `run_learning_curves.py`: Experience stratification
- `run_program_type_analysis.py`: Program type heterogeneity
- `run_disaster_context_analysis.py`: Disaster context heterogeneity
- `run_meta_analysis.py`: Summary statistics and forest plots

Reproducibility. All analyses are reproducible from the standardized panel data:

```
python src/pipeline.py ingest_data

python src/pipeline.py standardize_data

python src/pipeline.py build_panel

python src/pipeline.py build_features_std

python run_meta_analysis.py
```